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30 November 1992
SUPERSEDING
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MILITARY SPECIFICATION

ACTUATOR ASSEMBLY, ROTARY

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of rotary actuator assemblies unique to the suspension of the Armored Combat Earthmover, M9 ACE.

1.2 Classification. Suspension units shall be of the type specified (see 6.2):

Type I	- Rotary Actuator Assembly, Corner.
Type II	- Rotary Actuator Assembly, Intermediate.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2530

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SPECIFICATIONS
MILITARY

MIL-P-514	-Plate, Identification, Instruction and Marking, Blank.
MIL-L-2104	-Lubricating Oil, Internal - Combustion Engine, Tactical Service.
MIL-L-46167	-Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-C-46168	-Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-53039	-Coating Aliphatic Polyurethane, Single Components, Chemical Agent Resistant.

STANDARDS
MILITARY

MIL-STD-130	-Identification Marking of U.S. Military Property.
MIL-STD-193	-Painting Procedures and Marking for Vehicles, Construction Equipment and Material Handling Equipment.
MIL-STD-810	-Environmental Test Methods and Engineering Guidelines.

2.1.2 Other Government documents, drawings and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS
ARMY

12325801	-Roadarm.
12331881	-Seal, Shaft.
12335056	-Retainer, Roadarm.
12335135	-Plug, Special Long.
12335151	-Flange, Index.
12335160	-Ring, Backup.
13204E2560	-Rotary Actuator Assembly, Corner.
13204E2561	-Rotary Actuator Assembly Intermediate.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issue of the document which is indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J1165 -Reporting Cleanliness Levels of Hydraulic Fluids.

(Applications for copies of SAE publications should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified, the contractor shall furnish actuator assemblies which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the assembly to be furnished to the Government. All subsequent assemblies delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes. Any change in production methods after approval shall require a first article resubmittal.

3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards, and drawings. Material shall be free of defects which adversely affect performance or serviceability of the finished product.

3.2.1 Qualified products. The contractor shall be responsible for using parts and assemblies from Qualified Products Lists (QPLs) whenever available. Contractor's inspection records shall specifically list all QPL items by number and date of QPL, name of supplier and part or drawing number(s). When parts and assemblies are approved as qualified products but not listed on the QPL, the contractor shall list the product by number and date of the approved document and name of supplier(s).

3.2.2 Hydraulic oil. Hydraulic oil used in the actuator assemblies shall conform to grade 10 of MIL-L-2104 for ambient temperatures of +40 °F (+4 °C) and above, and to MIL-L-46167 for

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-15 °F (-26 °C) and below. Either of these oils can be used at ambient temperatures between +40 °F (+4 °C) and -15 °F (-26 °C).

3.2.3 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.3.2).

3.3 Construction. Actuator assemblies shall conform to the appropriate drawing listed in 2.1.2, and type specified in 1.2.

3.3.1 Interchangeability of parts. Component parts and assemblies of each actuator assembly shall be so constructed that any part(s), may be installed in a like assembly without modification.

3.4 Performance.

3.4.1 General performance - Type I and Type II actuator assemblies.

3.4.1.1 Proof pressure. The actuator assembly shall withstand testing per 4.6.1 without external leakage or deformation and shall subsequently meet all requirements of 3.4.1.2 through 3.4.4.

3.4.1.2 General external leakage. When tested as specified in 4.6.2, there shall be no visible external leakage.

3.4.1.3 Shaft seal leakage. There shall be no visible external leakage past shaft seal 12331881 when tested per 4.6.3.

3.4.1.4 Intentionally left blank.

3.4.1.5 Total shaft rotation. When tested per 4.6.5, total shaft rotation shall be 72 degrees plus or minus 1 degree.

3.4.1.6 Breakout pressure. The pressure required to initiate shaft rotation shall not exceed 300 psi (2070 kPa) and operation shall be smooth over total rotation range when tested per 4.6.6.

3.4.2 Performance - Type I actuator assembly.

3.4.2.1 Internal leakage. Internal leakage shall not exceed the rates specified in table I (see 4.6.7, 4.6.9 thru 4.6.15, 4.6.17, and 4.6.18).

3.4.2.2 Relief valve setting. Relief valve shall open at 4200 ± 100 psi (19 Lpm minimum flow at 28,960 ± 690 kPa), when tested per 4.6.16.

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TABLE I. Internal leakage - Type I actuator assembly.

No	Test item	Leakage oz/min	Requirement* (cu cm/min)	Test Requirement
1.	Vane seal	10.14	(300)	4.6.7.1
2.	Piston ring	10.14	(300)	4.6.7.2
3.	Wheel valve:			
3a.	Port 11 O-rings (2 at each end)	0.17	(5)	4.6.9
3b.	4-7A wheel valve leakage & related O-ring	0.17	(5)	4.6.10
3c.	1-2 wheel valve leakage & related O-ring	0.17	(5)	4.6.10
3d.	Port 3 wheel valve O-rings (2)	0.17	(5)	4.6.12
3e.	Port 1 wheel valve spool O-ring	0.17	(5)	4.6.13
3f.	2-3 wheel valve leakage & related O-ring	0.17	(5)	4.6.14
3g.	Port 4 wheel valve O-rings (2)	0.17	(5)	4.6.14
4.	Shock check valve	1.01	(30)	4.6.10
5.	Leveling valve seats:			
5a.	"Fill" valve seat	0.34	(10)	4.6.11
5b.	"Drain" valve seat	0.34	(10)	4.6.14
6.	Leveling valve stems:			
6a.	"Fill" valve stem	1.69	(50)	4.6.14
6b.	"Drain" valve stem	1.69	(50)	4.6.10
7.	Combined leakage (total of 1, 2, 6a, and 6b)	16.90	(500)	4.6.15
8.	Relief valve:			
8a.	Internal leakage	1.01	(30)	4.6.17
8b.	Reseat leakage	1.01	(30)	4.6.18

*maximum allowed ounces per minute, oz/min (cubic centimeters per minute, cu cm/min).

3.4.2.3 Leveling valves.

3.4.2.3.1 Leveling "fill" valve closing. The leveling "fill" valve shall close within the rotation specified in table II when tested per 4.6.19 through 4.6.22.

3.4.2.3.2 Leveling "drain" valve opening. The leveling "drain" valve shall open within the rotation specified in table II when tested per 4.6.19 through 4.6.22.

TABLE II. Leveling valve operating positions.

Leveling Valve Cam Position (Cam Setting)	Fill Valve Closing (Amount of Shaft Rotation from a fully Rotated Position)	Drain Valve Opening (Additional Rotation From when Fill Valve Closes)
1R	59° ± 3° CW from full CCW	6° to 12°
1L	59° ± 3° CCW from full CW	6° to 12°
4R	46° ± 3° CCW from full CW	6° to 12°
4L	46° ± 3° CW from full CCW	6° to 12°

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3.4.3 Performance - Type II actuator assembly.

3.4.3.1 Internal leakage. Leakage shall not exceed the rates specified in table III (see 4.6.7 and 4.6.8).

TABLE III. Internal leakage - Type II actuator assembly.

No	Test item	Leakage oz/min	Requirement* (cu cm/min)	Test Requirement
1.	Vane seal	10.14	(300)	4.6.7
2.	Piston ring	10.14	(300)	4.6.7
3.	Combined leakage (total of 1 & 2)	16.90	(500)	4.6.8

*maximum allowed ounces per minute, oz/min (cubic centimeters per minute, cu cm/min).

3.4.4 Environmental. Actuator assemblies shall be capable of operating in ambient temperatures of -50 °F to 120 °F (-46 °C to 49 °C) and with oil temperatures of -50 °F to 250 °F (-46 °C to 121 °C), without affecting their ability to meet the other requirements of 3.4. When prepared for storage, the actuator shall withstand temperature extremes of 160 °F to -50 °F without deterioration of components that prevent proper functioning of the unit (see 4.6.23).

3.5 Identification and marking. Unless otherwise specified on the applicable drawing, each actuator assembly and each major component shall be marked in accordance with MIL-STD-130.

3.6 Nameplate. Unless otherwise specified on the applicable drawing, nameplate, data and instruction plates shall conform to the applicable provisions of MIL-P-514.

3.7 Finish. Unless otherwise specified (see 6.2), actuator assemblies shall be cleaned, treated, primed and painted in accordance with the drawing.

3.8 Workmanship. Workmanship shall be of a quality to assure that actuator assemblies and components are free from defects of a workmanship nature.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection and test requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection and test requirements specified herein, unless disapproved by the Government. The Government reserves

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the right to perform or witness any of the inspections or tests set forth in the specification where deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment. Unless otherwise specified in the contract, the contractor is responsible for the provision and maintenance of all inspection and test equipment necessary to assure that the supplies and services conform to contract requirements. Commercial, modified commercial, or contractor designed equipment or measuring set-ups shall be capable of repetitive measurements to an accuracy of 10 percent of the component tolerance.

4.1.2 Materials and construction. To determine conformance to 3.2 through 3.3.1, inspection and material certification records shall be maintained by the contractor. Records shall be subject to review by the Government and shall include date, part, or characteristic identification, inspection results, and disposition of lot (accepted or rejected). Corrective action taken on noted defects shall be subject to approval by the Government.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
    1. Examination (see 4.5.2).
    2. Acceptance tests (see 4.5.3.1).
    3. Control tests (see 4.5.3.2).

4.3 Inspection conditions. All inspections and tests shall be conducted under the following conditions:

a. Hydraulic fluid shall conform to grade 10 of MIL-L-2104, and be maintained at a temperature of 90 °F ± 20 °F (32 °C ± 11 °C), except when otherwise specified.

b. Cleanliness of fluid shall conform to ISO code 19/16 as specified in SAE J1165.

4.3.1 Test equipment. The following test equipment shall be available for performance of the tests and inspections specified herein:

a. Fixture(s) to securely hold and position an actuator under test. It shall be capable of Restricting the movement of the shaft when test conditions require.

b. Hydraulic power source(s) capable of providing a minimum of 8000 psi (55160 kPa) pressure at 1 gpm (3.8 Lpm) flow and 4500 psi (31030 kPa) pressure at 5 gpm (19 Lpm) flow.

c. Gages to measure pressure and beakers to measure leakage. The gages shall be accurate within plus or minus 2 percent of full scale, and the beakers within plus or minus 5 percent of reading.

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4.3.2 Internal fluid leakage. Internal leakage shall be interpreted as any flow coming out of the port or ports designated in the test procedures of 4.6. The leakage rate shall stabilize for 30 seconds prior to measuring leakage.

4.3.3 External fluid leakage. External leakage shall be interpreted as any leakage to the outside of the actuator assembly (see 3.4.1.2 and 3.4.1.3).

4.3.4 Preparation for testing. The following procedures shall be accomplished prior to the starting of tests:

4.3.4.1 Establishing direction of shaft rotation. Except where otherwise specified in 4.6, the procedures of 4.6 are based upon applying pressure to face B of type I actuators (face A ports plugged), and to face A of type II actuators (face b ports plugged), so that application of pressure to the unplugged port 2 will result in clockwise rotation of shaft as viewed from the shaft end. The ports in the opposite face may be used instead, but that will result in reversed direction of rotation. Looking at the shaft end of the actuator assembly, face A is the right hand side, and face B is the left hand side. (See figures 1 and 2 for face and port locations.)

4.3.4.2 Plugged ports (see figures 1 and 2).

a. Type I actuator. The two ports No. 1, and all unnumbered ports are to be plugged for all tests. The four ports in face A are to be plugged for all tests except when otherwise specified. All other numbered ports are to remain OPEN unless otherwise specified.

b. Type II actuator. The two ports in face B are to be plugged for all tests.

c. Port 7B. Port 7B in both type I and type II actuators shall remain open for all tests except when pressure is applied to port 7B (DO NOT PLUG PORT 7B).

4.3.4.3 Check long plug (type I actuator only). Inspect long plug 12335135 in port 2 of face A to be sure it is a special long plug with seal in groove near inner end of plug.

4.3.4.4 Torque plugs. Tighten plugs to the torques specified on drawings 13204E2560 and 13204E2561.

4.3.4.5 Check cam "V" marks (type I actuator only). Remove socket head screw that secures index flange 12335151 to cam shaft at end of main shaft. Inspect to be sure that the "V" mark on the index flange is properly aligned (not 180° out of position). Inspect by removing the flange to be sure the "V" mark on flange is aligned over "V" at bottom of one slot in cam shaft.

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4.3.4.6 Check cam setting (type I actuator only). With index flange properly oriented per 4.3.4.5, the "V" mark on flange should point to index mark 1R or 4L stamped on shaft. Reset cam position if required. Reinstall socket head screw.

4.3.4.7 Install and retain shaft seal. Install shaft seal 12331881, ring 12335160, and packing MS28775-427 over end of shaft. Retain these items with roadarm 12325801 or suitable collar, and retain collar with retainer 12335056 or suitable equivalent.

4.3.4.8 Intentionally left blank.

4.3.4.9 Fill drain chamber with oil.

a. Type I actuator. With the "Fill" leveling valve cap, spring, and stem removed, add oil into port 7B until oil flows out the "fill" valve port. Reinstall valve parts, and torque cap.

b. Type II actuator. Add oil into port 7B to fill drain chamber to the level of port 7B.

4.3.4.10 Fill vane chambers with oil. Alternately apply low pressure to ports 2 and 17 of type I actuator, or to ports 2 and 7 of type II actuator, and rotate shaft back and forth through several full cycles.

4.3.4.11 Intentionally left blank.

4.4 First article inspection. Unless otherwise specified (see 6.2), the contractor shall furnish one type I and one type II actuator assembly, from the first ten units of each type produced under the production contract, for first article inspection. First article samples shall be examined as specified in table V and tested as specified in table VI. Approval of the first article samples by the Government shall not relieve the contractor of his obligation to supply actuator assemblies that are fully representative of those inspected as first article samples. Any changes or deviation of the production units from the first article samples shall be subject to the approval of the contracting officer.

4.4.1 First article inspection failure. Failure of any first article sample to pass specified examinations or tests shall be cause for refusal to grant first article approval and to accept production units until corrective action by the contractor has been approved by the Government.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all actuator assemblies of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

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4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with table IV. Accept on zero defects, and reject on one or more defects.

TABLE IV. Sampling of for examination.

Lot size	Sample Size	
	For Major Characteristics	For Minor Characteristics
2 to 13	Entire Lot	3
14 to 25	13	3
26 to 50	13	5
51 to 90	13	6
91 to 150	13	7
151 to 280	20	10
281 to 500	29	11
501 to 1200	34	15
1201 to 3200	42	18
3201 to 10,000	50	22

NOTE: Accept lot on zero defects. Reject lot on one or more defects.

4.5.2 Quality conformance examinations.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined for the defects listed in table V. Accept on zero defects. Reject on one or more defects.

TABLE V. Classification of defects.

Category	Defect	Method of examination
Major		
101	Construction: Not to drawing (see 3.3).	Visual
102	Dimensions affecting interchangeability: Not within tolerance (see 3.3.1).	Visual and SIE 1/
103	Workmanship: Not as specified (see 3.8).	Visual
Minor		
201	Dimensions not affecting interchangeability: Not to tolerance (see 3.3.1).	Visual and SIE 1/
202	Identification and marking: Not as specified (see 3.5).	Visual
203	Nameplates: Not as specified (see 3.6).	Visual
204	Painting: Not as specified, poor coverage (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment

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4.5.3 Quality conformance tests.

4.5.3.1 Acceptance tests (100 percent). Each actuator assembly shall be subjected to the tests specified in table VI.

4.5.3.1.1 Failure. Actuator assemblies that fail to pass any of the acceptance tests specified in table VI may be repaired and resubmitted for retest. The retest, as a minimum shall consist of the failed tests and any other tests as deemed necessary by the Government inspector.

4.5.3.2 Control tests. Control test samples shall be selected from a production lot which has passed quality conformance examinations specified in 4.5.2 and acceptance tests specified in 4.5.3.1. Control test samples shall be selected at the rate of one from each lot of 20 actuator assemblies consecutively produced, except that not more than one shall be tested in a 2 week period, nor less than one in a 12 month period. The control test sample shall be subjected to the tests specified in table VI.

4.5.3.2.1 Failure. Failure of a sample to pass control testing shall be cause for the Government refusal to accept actuator assemblies until corrective action taken by the contractor has been approved by the Government.

TABLE VI. Classification of tests.

Title	Requirement	Tests	First article	Quality conformance	
				Acceptance (100%)	Control
General Performance (Types I and II)	3.4.1				
Proof pressure	3.4.1.1	4.6.1	X		X
General external leakage	3.4.1.2	4.6.2	X	X	
Shaft seal leakage	3.4.1.3	4.6.3	X	X	
Total shaft rotation	3.4.1.5	4.6.5	X	X	
Breakout pressure	3.4.1.6	4.6.6	X	X	
Performance (Type I)	Table I item no.				
Vane seal leakage	1	4.6.7.1	X	X	
Piston ring leakage	2	4.6.7.2	X	X	
Port 11 wheel valve O-ring leakage	3a	4.6.9	X	X	
4-7A wheel valve 1 kg	3b	4.6.10	X	X	
1-2 wheel valve 1 kg	3c	4.6.10	X	X	
Port 3 wheel valve O-ring leakage	3d	4.6.12	X	X	

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TABLE VI. Classification of tests. - cont.

Title	Requirement	Tests	First article	Quality conformance	
				Acceptance (100%)	Control
Port 1 wheel valve O-ring leakage	3e	4.6.13	X	X	
2-3 wheel valve 1 kg	3f	4.6.14	X	X	
Port 4 wheel valve O-ring leakage	3g	3.6.14	X	X	
Shock check valve 1 kg	4	4.6.10	X	X	
Level "Fill" valve seat leakage	5a	4.6.11	X	X	
Level "Drain" valve seat leakage	5b	4.6.14	X	X	
Level "Fill" valve stem leakage	6a	4.6.14	X	X	
Level "Drain" valve stem leakage	6b	4.6.10	X	X	
Combined leakage	7	4.6.15	X	X	
Relief valve leakage	8a	4.6.17	X	X	
Relief valve reseat leakage	8b	4.6.18	X	X	
Relief valve setting	3.4.2.2	4.6.16	X	X	
Leveling "Fill" valve close	3.4.2.3.1	4.6.19 – 22	X	X	
Leveling "Drain" valve open	3.4.2.3.2	4.6.19 – 22	X	X	
Performance (Type II)	Table III Item no.				
Vane seal leakage	1	4.6.7.1	X	X	
Piston ring leakage	2	4.6.7.2	X	X	
Combined leakage	3	4.6.8	X	X	
Temperature extremes performance	3.4.4	4.6.23	X		

4.6 Methods of inspection. The following test procedures, or equivalent procedures acceptable to the Government, shall be applied to determine conformance to requirements of 3.4. Prior to performing the following tests, preparation for testing in accordance with 4.3.4 shall have been accomplished to provide a proper setup.

4.6.1 Proof pressure. To determine conformance to 3.4.1.1:

a. Position shaft at midtravel position (36° ± 20° rotation form full rotation position).

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b. For type I actuator, remove relief valve and plug all ports except 7B, 11, 2, and 17. (Be sure Port 7B is not plugged.)

c. For type II actuator, be sure port 7B is not plugged.

d. Apply 7500 to 8000 psi (51,700 to 55,100 kPa) to ports 2 and 17 simultaneously for type I actuator, or to ports 2 and 7 simultaneously for type II actuator, and hold this pressure for 30 seconds.

4.6.2 General external leakage. To determine conformance to 3.4.1.2:

a. Position shaft at midtravel position (36° ± 10° rotation from full rotation position).

b. For type I actuators, install relief valve (if removed for 4.6.1), install a "bleed" type plug in port 7, and plug all ports except 7B, 11, 2, and 17 (be sure port 7B is not plugged).

c. For type II actuators, be sure port 7B is not plugged.

d. Apply 6200 ± 200 psi (42,749 ± 1379 kPa) to ports 2 and 17 simultaneously for type I actuator, or to ports 2 and 7 for type II actuator, hold this pressure for 2 minutes, and inspect for external leakage (flow from port 11 is external leakage, but flow from port 7B is not).

4.6.3 Shaft seal leakage. To determine conformance to 3.4.1.3:

a. For type I actuator, plug ports 2, 8, 9, 14, and 7A (other numbered ports may be open or plugged).

b. Apply 20 ± 5 psi (138 ± 35 kPa) to port 7B and hold this pressure for 5 minutes.

Inspect for external leakage past the shaft seal.

4.6.4 Intentionally left blank.

4.6.5 Total shaft rotation. To determine conformance to 3.4.1.5:

a. Apply low pressure (approximately 500 psi) to port 17 of type I actuator, or to port 7 of type II actuator, and rotate shaft to full counterclockwise position at a rotation rate of 7 ± 6 per second. Apply low pressure to port 2 to rotate to full clockwise at the same rate of rotation.

b. Hold 200 psi (1379 kPa) to maintain position. Calculate the total shaft rotation between full CW and full CCW.

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4.6.6 Breakout pressure. To determine conformance to 3.4.1.6:

a. For type I actuator only:

1. Plug port 3.
2. Apply 3200 ± 200 psi (22,064 ± 1,380 kPa) to port 11 and hold this pressure.

b. Slowly apply pressure to the appropriate port (2 or 17) and increase pressure until wing shaft rotation occurs and continues. Perform this operation with wing shaft in the full CW position, the full CCW position and in both directions from mid-travel. Record the pressure required to initiate rotation in each case.

4.6.7 Vane seal and piston ring leakage. To determine conformance to items 1 and 2 of tables I and III:

4.6.7.1 Vane seal leakage.

a. For type I actuator:

1. Plug port 3.
2. Apply 3200 ± 200 (22,064 ± 1,380 kPa) to port 11 and hold this pressure.

b. Alternately apply pressure to ports 2 and 17 for type I actuator, or to ports 2 and 7 for type II actuator, and cycle shaft 15 times in each direction of rotation. Apply 3200 ± 200 psi (22,064 ± 1,380 kPa) for 5 seconds at each end of cycle (at full rotation).

c. Apply 3200 ± 200 psi (22,064 ± 1,380 kPa) to the ports specified in table VII, with the shaft secured at the positions listed in table VII. In each case, hold 3200 ± 200 (22,064 ± 1,380 kPa) and measure leakage form the ports specified below (see 4.3.2).

1. Type I actuator vane seal leakage from port 2 or 17, whichever is not used to apply pressure.
2. Type II actuator vane seal leakage from port 2 or 7, whichever is not used to apply pressure.

4.6.7.2 Piston ring leakage.

a. For type I only:

1. Plug port 3.
2. Apply 3200 ± 200 psi (22,064 ± 1,380 kPa) to port 11 and hold this pressure.

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b. Alternately apply pressure to ports 2 and 17 for type I actuator or to ports 2 and 7 for type II actuator, and cycle shaft 15 times in each direction of rotation. Apply 3200 ± 200 psi (22,064 ± 1,380 kPa) for 5 seconds at each end of cycle (at full rotation).

c. Apply 3200 ± 200 psi (22,064 ± 1,380 kPa) to port 2, with the shaft secured in mid position. Hold 3200 ± 200 psi (22,064 ± 1,380 kPa) and measure leakage from port 7b (see 4.3.2).

4.6.8 Combined leakage, type II actuator. To determine conformance to item 3 of table III, calculate the total of the two leakage rates measured in 4.6.7.1 and 4.6.7.2.

4.6.9 Port 11 wheel valve O-ring leakage. To determine conformance to item 3a of table I:

a. Apply 3200 ± 200 (22,064 ± 1,380 kPa) to port 11 and hold this pressure.

b. Measure leakage at ports 7A and 7B (see 4.3.2).

TABLE VII. Wingshaft Rotation Positions for Vane Seal Leakage Tests and for Breakout PressureTests.

	Shaft Position (CCW Rotation Angle from Full CW)	Port Appplied with Pressure
Type I Actuators corner	5 ° ± 5 ° 36 ° ± 5 ° 36 ° ± 5 ° 67 ° ± 5 °	Port 2 side B Port 2 side B Port 17 side B Port 17 side B
Type II Actuators (Intermediate)	5 ° ± 5 ° 36 ° ± 5 ° 36 ° ± 5 ° 67 ° ± 5 °	Port 2 side A Port 2 side A Port 7 side A Port 7 side A

4.6.10 4-7A Wheel valve leakage, 1-2 wheel valve leakage, shock check valve leakage and leveling “drain” valve stem leakage. To determine conformance to items 3b, 3c, 4, and 6b of table I:

a. Apply 3200 ± 200 psi (22,064 ± 1,380 kPa) to port 11 and hold this pressure.

b. Plug port 3.

c. Apply low pressure to port 2 and rotate shaft to full clockwise position.

d. Remove plug from port 3.

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e. Apply 3200 ± 200 psi (22,064 ± 1,380 kPa) to port 14, and hold this pressure.

f. Measure leakage at the following ports (see 4.3.2):

1. Leakage from port 7A (4-7A wheel valve leakage).
2. Leakage from port 3 (1-2 wheel valve leakage).
3. Leakage from port 8 (shock check valve leakage).
4. Leakage from port 7B (leveling "Drain" valve stem leakage).

NOTE: If no leakage occurs from port 7B, the drain chamber was not properly filled prior to test. Fill per 4.3.4.9.a and retest.

4.6.11 Leveling “fill” valve seat leakage. To determine conformance to item 5a of table I:

a. If continuing from 4.6.10, release pressure from port 14.

b. If not continuing from 4.6.10, apply procedures a, b, and c, of 4.6.10.

c. Apply 3200 ± 200 psi (22,064 ± 1,380 kPa) to port 9 and hold this pressure.

d. Measure leakage from port 8 (see 4.3.2).

4.6.12 Port 3 wheel valve O-ring leakage. To determine conformance to item 3d of table I:

a. If continuing from 4.6.11, release pressure from port 9, and remove plug from port 3.

b. Apply 3200 ± 200 psi (22,064 ± 1,380 kPa) to port 11 and hold this pressure.

c. Install special “long” plug 12335135 into port 2 of face B.

d. Apply 3200 ± 200 psi (22,064 ± 1,380 kPa) to port 3 and hold this pressure.

e. Measure leakage from port 7B (see 4.3.2).

4.6.13 Port 1 wheel valve spool O-ring leakage. To determine conformance to item 3e of table I:

a. If continuing from 4.6.12, release pressure from ports 3 and 11, and remove “long” plug from port 2 of face B.

b. Install shutoff valve at port 17.

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c. Apply low pressure to port 17 and rotate shaft to full CCW position. Close shutoff valve to hold shaft position.

d. Plug port 8.

e. Hold 3200 ± 200 psi (22,064 ± 1,380 kPa) at port 14 and measure leakage from port 7B (see 4.3.2).

4.6.14 2-3 wheel valve leakage, port 4 wheel valve O-rinq leakage leveling "drain" valve seat leakage and leveling "fill" valve stem leakage. To determine conformance to items 3f, 3g, 5b, and 6a of table I:

a. If continuing from 4.6.13, maintain full CCW shaft rotation with shutoff valve closed at port 17, retain "long" plug in port 2, and maintain zero pressure at port 11. Release pressure from port 14.

b. If not continuing from 4.6.13, apply procedures b, c, and d of 4.6.13.

c. Plug ports 8 and 14, and open ports 3 and 7A.

d. Apply 3200 ± 200 (22,064 ± 1,380 kPa) to port 9 and hold this pressure. Measure leakage at the following ports (see 4.3.2):

1. Leakage from port 3 (2-3 wheel valve leakage).
2. Leakage from port 7A (port 4 wheel valve O-ring leakage plus leveling "drain" valve seat leakage.)
3. Leakage from port 7B (leveling "fill" valve stem leakage).

NOTE: If no leakage occurs from port 7B, the drain chamber was not properly filled prior to test. Fill per 4.3.4.9.a and retest.

4.6.15 Combined leakage, type I actuator. To determine conformance to item 7 of table I, calculate the total of the four leakage rates measured for items 1, 2, 6a, and 6b of table I.

4.6.16 Relief valve setting. To determine conformance to 3.4.2.2:

a. Apply 3200 ± 200 psi (22,064 ± 1,380 kPa) to port 11 and hold this pressure.

b. Plug port 3.

c. Apply pressure at port 2. Slowly increase the port 2 pressure and record the pressure reading when the relief valve opens. Reduce pressure to zero.

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4.6.17 Relief valve internal leakage. To determine conformance to item 8a of table I:

a. Maintain setup of 4.6.16 (apply 3200 ± 200 psi [22,064 ± 1,380 kPa] to port 11 and plug port 3).

b. Apply pressure to port 2. Slowly increase the pressure from 0 to 3750 psi (0 to 25,850 kPa), and maintain pressure at 3750 ± 200 psi (25,850 ± 1380 kPa). Decrease pressure to 0 psi. Measure leakage from port 7 (see 4.3.2).

c. Measure and record the maximum leakage rate from port 7 for port 2 pressures from 0 to 3750 psi (0 to 25,850 kPa).

4.6.18 Relief valve reseat leakage. To determine conformance to item 8b of table I:

a. Maintain setup of 4.6.16 and 4.6.17 (apply 3200 ± 200 psi [22,064 ± 1,380 kPa] to port 11 and plug port 3).

b. Apply pressure at port 2 and increase pressure to open the relief valve.

c. Slowly decrease pressure so valve closes. Continue to decrease pressure to 3200 psi (22,064 kPa), and hold pressure at 3200 ± 200 psi (22,064 ± 1,380 kPa). Measure leakage from port 7 (see 4.3.2).

4.6.19 Leveling "fill" valve closing and leveling "drain" valve opening, 1R position. To determine conformance to 3.4.2.3.1 and 3.4.2.3.2 for 1R position:

a. Maintain 0 pressure at port 11.

b. Set cam ar 1R position (requires full CW rotation; obtain by plugging ports 7A and 14, and applying pressure to port 2).

c. Apply low pressure to port 17 and rotate shaft to full CCW position.

d. Open port 7A, and plug ports 2, 8, and 14. Apply low pressure to port 9 and SLOWLY rotate shaft clockwise until the leveling "fill" valve closes (rotation ceases). Measure this rotation angle.

e. Open port 9 and apply low pressure to port 14 at a very low flow rate to resume SLOW clockwise rotation until the leveling "drain" valve opens (rotation again ceases, and the oil fed into port 14 comes out port 7A). Measure this rotation angle.

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4.6.20 Leveling "fill" valve closing and leveling "drain" valve opening, 4L position. To determine conformance to 3.4.2.3.1 and 3.4.2.3.2 for 4L position, apply the procedures of 4.6.19 with the following exceptions:

- If continuing from 4.6.19, it is not necessary to repeat part b.
- In part c, set cam at 4L position instead of 1R position.

4.6.21 Leveling "fill" valve closing and leveling "drain" valve opening, 1L position. To determine conformance to 3.4.2.3.1 and 3.4.2.3.2 for 1L position:

a. Remove the plugs from the four ports in face A (ports 2, 3, 11, and 17), and install them into the four ports of face B.

b. Maintain 0 pressure at port 11 of face A.

c. If not continuing from 4.6.20, apply the procedures of 4.3.4.11 to insure that the wheel valve is in "sprung" position.

d. Set cam at 1L position (requires full CW rotation; obtain by applying pressure to port 17 of face A).

e. Open port 7A, and plug ports 2, 8, and 14. Apply low pressure to port 9 and SLOWLY rotate shaft CCW until the level "fill" valve closes (rotation ceases). Measure this rotation angle.

f. Open port 9 and apply low pressure to port 14 at a very low flow rate to resume SLOW CCW rotation until the leveling "drain" valve opens (rotation again ceases, and the oil fed into port 14 comes out port 7A). Measure this rotation angle.

4.6.22 Leveling "fill" valve closing and leveling "drain" valve 4R position. To determine conformance to 3.4.2.3.1 and 3.4.2.3.2 for 4R position, apply the procedures of 4.6.21 with the following exceptions:

- If continuing from 4.6.21, it is not necessary to repeat part c.
- In part d, set cam at 4R position instead of at 1L position.

4.6.23 Environmental. To determine conformance to 3.4.4, the actuator shall meet all performance requirements specified in 3.4.4 before during and immediately after exposure to the environmental conditions specified herein and shall be tested in accordance with the requirements of MIL-STD-810, method 501.2, procedures I and II and method 502.2, procedures I and II.

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4.6.23.1 High temperature. Method 501.2. The actuator shall be tested as follows:

a. Procedure I. Storage.

b. Procedure II. Operation.

STEP 1. Place the actuator in a test chamber for not less than 6 hours (h) at a temperature of 120 °F (49 °C).
STEP 2. Raise the chamber temperature to 160 °F (71 °C) within a time period of 1 h and then maintain at that temperature for an additional 4 h. Lower the chamber temperature back to 120 °F (49 °C) within a time period of 1 h.
STEP 3. Repeat steps 1 and 2 two additional times (making a total of three 12 h cycles).
STEP 4. Adjust the chamber temperature to 120 °F (49 °C), and maintain until temperature stabilization of the actuator is reached.
STEP 5. Operate the actuator until it is stabilized, and obtain results.
STEP 6. Return the actuator, nonoperating, to standard ambient conditions and stabilize.

After the test, the actuator shall be checked for any damage.

4.6.23.2 Low temperature. Method 502.2. The actuator shall be tested as follows:

a. Procedure I. Storage.

b. Procedure II. Operation.

STEP 1. Place the actuator in a test chamber for not less than 24 h at a temperature of -50 °F (-46 °C). Raise the chamber temperature to -25 °F (-32 °C) and maintain until temperature stabilization -25 °F (-32 °C) of the actuator is reached.
STEP 2. Operate the actuator until it is stabilized, without external aids and with provisions for oil cooling, and obtain results.
STEP 3. Return the actuator, nonoperating, to standard ambient conditions and stabilize.
STEP 4. The rate of temperature change (steps 1, 2 and 3) may be the maximum attainable by the chamber but shall not be greater than 18 °F (10 °C) per minute.

After the tests, the actuator shall be checked for any damage or malfunction.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The actuator assemblies covered by this specification are intended for use as components of the suspension system of the Armored Combat Earthmover M9 ACE.

6.2 Ordering data. Acquisition documents must specify the following:

a. Type, number, and date of this specification.
b. Type(s) of actuator assembly(s) to be furnished (see 1.2).
c. If the finish shall be other than as specified (3.7).
d. If first article inspection is not required (see 4.4).
e. Selection of applicable level and packaging requirements (see 5.1).

6.3 Supplementary engineering data.

6.3.1 Oil displacement (approximate).

0.86 cubic inches per degree of shaft rotation (62 cubic inches in the 72 full rotation).

6.3.2 Output torque (theoretical).

45,000 pound-inches per 1000 psi input pressure.

6.3.3 Travel buffering range.

2.5 approximately at each end of the 72 full shaft rotation.

6.3.4 Weight (approximate).

a. Type I actuator - 260 lb (dry).
b. Type II actuator - 225 lb (dry).

6.4 Definitions.

6.4.1 Leakage.

a. Weep - any evidence of fluid beyond a seal or joint.
b. Seep - any evidence of fluid beyond a seal or joint that does not result in the formation of a droplet.
c. Leak - any evidence of fluid beyond a seal or joint that results in the formation of droplet.

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d. Drip - any evidence of fluid beyond a seal or joint where droplets form and fall.

6.4.2 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.3.3).

6.4.3 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural , operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.5 Subject term (key word) listing.

Assembly, rotary actuator.
Rotary actuator assembly.
Suspension system rotary actuator.

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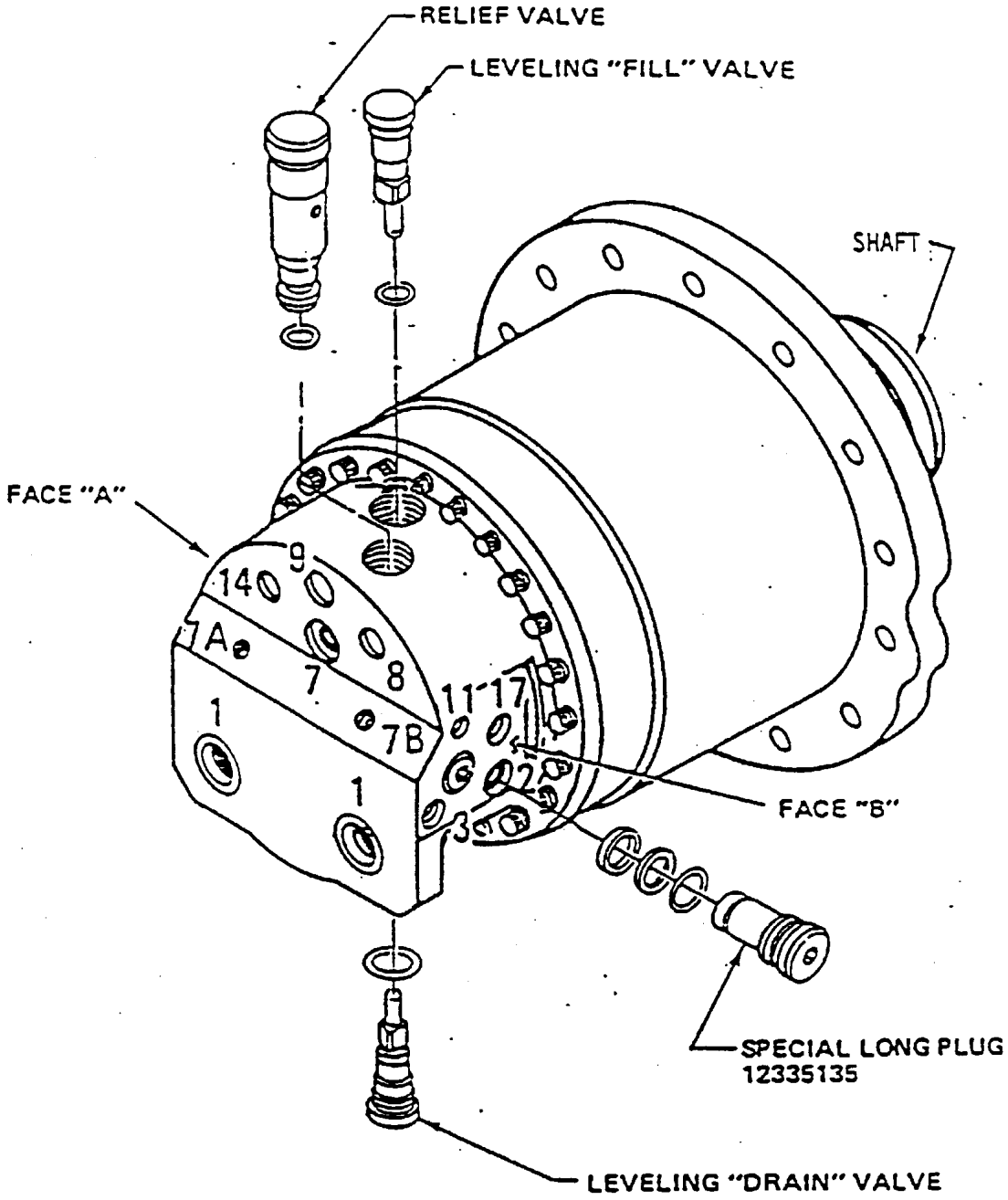


FIGURE 1. Type I Actuator

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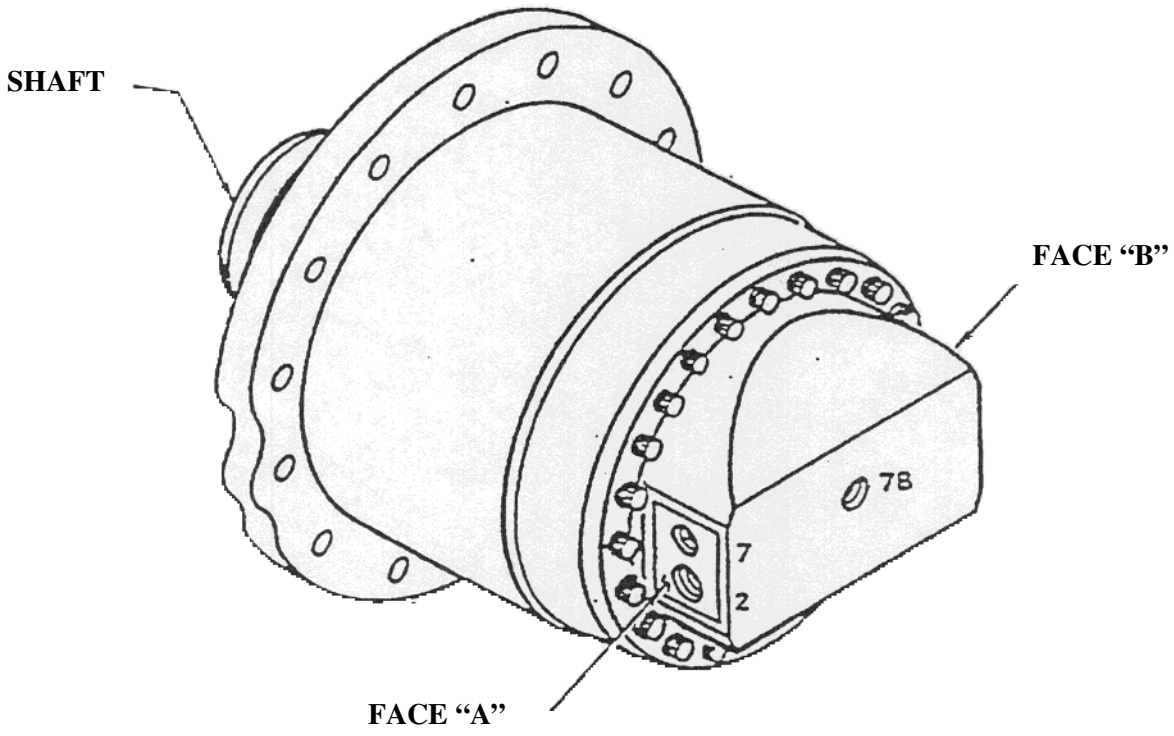


FIGURE 2. Type II Actuator

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Custodian:
Army - AT

Preparing activity:
Army - AT

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